Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 10.1 · Convert and Measure

The trigonometric circle is used to measure radian angles.

1. Convert 150 $^\circ$ and -90° in radians, showing the conversion factor.
2. A circular sector has a radius of 4 cm and an angle of $\frac{\pi}{3}$ radians. Calculate the exact length of his bow.
3. Explain why a angle measurement can be negative while a geometric arc length is positive.

Exercise 10.2 · Reduce angles

We consider the real $\frac{17\pi}{6}$, $-\frac{7\pi}{4}$ and 5π.

1. For each, determine a representative in [0; 2 π[having the same image on the circle Trigonometric. Show the number of turns added or removed.
2. Place the three images on a circle and identify them by the actual starting.

Exercise 10.3 · Accurate values

The exact values of the cosine and sine of $\frac{2\pi}{3}$, $-\frac{\pi}{6}$ and $\frac{5\pi}{4}$.

1. Place the three images on the trigonometric circle. For each, specify the expected sign of its two coordinates.
2. Calculate the exact values using a remarkable angle and symmetry. answer in a table: angle, cosine, sine.

Exercise 10.4 · Find the sign

A real one x checks $\sin x = \frac{5}{13}$. Its image on the trigonometric circle is located in the second quadrant, therefore to the left of the axis of the y-axis and above the axis of the abscisses.

1. Determine the exact value of cos x. Justify both its absolute value and its sign.

2. Explain why the only sin data x would not have been enough to choose the sign of $\cos x$.

Exercise 10.5 · One wheel

One wheel of radius 0,30 m rolls on a horizontal ground without slipping. It rotates angle $\frac{5\pi}{3}$ The distance travelled by its axis in the direction of displacement is sought.

1. Calculate this distance in exact form by explaining the link with the length of the arc.

2. Explain why one should not use the length of a rope in the circle.
fixed point on the edge of the wheel, seen from the ground, is it that of the axis?

Exercise 10.6 · String and scalar product

On the centre trigonometric circle O, point A is the image of 0 and the point B that of $\frac{2\pi}{3}$.

1. Determine the exact coordinates of A and B, then calculate the length AB.

2. Retrieve this length by applying the relationship of Al-Kashi to the triangle OAB. Specify the lengths and angle used.

Exercise 10.7 · Read an equation on the circle

We're looking for real things. x of the interval [0; 2 π[whose image on the trigonometric circle has for abscisse $\frac{1}{2}$.

1. Translate the condition on the abscisse by an equation involving cos x.

2. Trace the corresponding vertical line on a circle, identify its intersections and give the
Two solutions. Check that they belong to the required interval.

Fiscal year 10.8 · Deferred review

The measuring angle shall be considered $\frac{25\pi}{4}$ The answers must be accurate, without approximation. decimal places.

1. Reduce this angle by showing an integer number of complete turns.

2. Deduce its cosine and sine. Justify the values using the trigonometric circle.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, exercise 32 p. 220, questions a and b. Use the exact value provided and angles partners, without calculator.

CQFD, Exercise 39 p. 220, Questions 1, 2, 3 and 5. Addressing the four issues of calculation and reading the circle; justify the answers on the copy.